

IAL STATISTICS 1 · RAPID REVISION SHEET

Pearson Edexcel WST01/01 · Distilled from 18 papers and mark schemes, October 2020 – Summer 2026

1 FIVE RULES THAT DECIDE YOUR GRADE

MARK-SCHEME RULES

1. Percentage-point z to **4 d.p.** (0.8416 not 0.84): separate B1.
2. **Write the standardisation** $\frac{x-\mu}{\sigma}$; calculator-only answers can score 0.
3. Every comment **in context**: name both variables (and units).
4. Regression: decimals, the question's letters, stated precision (46.0, not 46).
5. "Show that": full numerical expression, no wrong working, final line as printed.

Answers to 3 s.f. (awrt) unless told otherwise; keep full calculator values until the end.

2 DATA: QUANTILES, OUTLIERS, PLOTS

- **Raw data** (stem and leaf): median = $\frac{n+1}{2}$ th. Q_1 : $\frac{n}{4}$ not whole $\Rightarrow$ round up; whole $\Rightarrow$ average it and the next. Q_3 from $\frac{3n}{4}$.
- **Grouped data**: $\frac{n}{2}, \frac{n}{4}, \frac{3n}{4}$ with interpolation:

$$L + \frac{\text{pos} - \text{cf before}}{f} \times \text{width}$$
- **Outliers**: $< Q_1 - 1.5 \text{ IQR}$ or $> Q_3 + 1.5 \text{ IQR}$ (use the k given).
- **Box plot**: whiskers to last non-outliers (or to limits); outliers as $\times$.
- **Histogram**: area $\propto f$; f.d. = $f \div \text{width}$. Used for *continuous* data.
- **Bar scaling**: given bar $\rightarrow$ cm per unit (width) and cm^2 per item (area); height = area $\div$ width.
- **Boundaries**: "11–20 to nearest hour" = 10.5 to 20.5.

SKEWNESS

Positive (tail right)	$Q_3 - Q_2 > Q_2 - Q_1$; mean $>$ median
Negative (tail left)	$Q_3 - Q_2 < Q_2 - Q_1$; mean $<$ median
Symmetric	equal; mean $\approx$ median

Interpret: "more spread above the median". Skewed $\Rightarrow$ normal model unsuitable.

COMPARE TWO DISTRIBUTIONS (3 MARKS)

1. Compare **medians** (say "median").
2. Compare **IQR or range**.
3. Interpret one in context ("on average the Feb. weights were lower"). Skew/outlier comments score nothing.

3 MEAN, VARIANCE, CODING

FORMULAE

$$\bar{x} = \frac{\sum x}{n} \quad \sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2 = \frac{S_{xx}}{n}$$

Grouped: use $\sum f, \sum fx, \sum fx^2$ with mid-points.

- **Coding** $y = \frac{x-a}{b}$: $\bar{x} = a + b\bar{y}$, $\sigma_x = |b|\sigma_y$, $\sigma_x^2 = b^2\sigma_y^2$.
- Adding a constant: mean changes, spread does not (give that reason; "coding" alone is B0).
- **Combining groups**: add $n, \sum x, \sum x^2$ (use $\sum x^2 = n(\sigma^2 + \bar{x}^2)$).
- Adding a value equal to the mean: mean same, s.d. decreases.
- Grouped answers are *estimates* because the raw data are unknown.

4 CORRELATION AND REGRESSION

FORMULAE

$$S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n} \quad S_{xy} = \sum xy - \frac{\sum x \sum y}{n}$$

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} \quad b = \frac{S_{xy}}{S_{xx}} \quad a = \bar{y} - b\bar{x}$$

- $\sum y^2$ used in place of S_{yy} is M0.
- Line passes through $(\bar{x}, \bar{y})$; two regression lines meet there.
- x = explanatory (set/controlled); y = response.
- Linear coding: $|r|$ unchanged. Doubling x : r , intercept same; gradient halves.
- Adding the point $(\bar{x}, \bar{y})$: r unchanged.

WORDING THAT SCORES

r : "As height increases, weight tends to increase."
Linear? " r close to 1, points close to a straight line."
Gradient: "Each extra 1 cm of height adds 3.27 kg (on average)."
Reliability: "153 is within the data's heights (138–157): interpolation, reliable"; outside $\Rightarrow$ extrapolation. Must quote the x -range.
Why models: simplify/understand the real world; can be refined (not cost/time alone).

5 PROBABILITY

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A | B) = \frac{P(A \cap B)}{P(B)}$ ("given" $\Rightarrow$ divide)
- **Independent:** $P(A \cap B) = P(A)P(B)$ or $P(A | B) = P(A)$. Show numbers *and* conclude in words.

- **Mutually exclusive:** $P(A \cap B) = 0$; separate circles. $P(C) + P(D) > 1 \Rightarrow$ cannot be M.E.
- **Venn with unknowns:** total = 1, then one equation per condition. Fill every region, incl. outside.
- **Survey Venn:** start from the centre (all three), work outwards.
- **Tree:** multiply along, add across. Reverse conditional = path $\div$ all paths meeting the condition.
- **Without replacement:** denominators drop by 1; quadratic in n , reject invalid roots.

6 DISCRETE RANDOM VARIABLES

FORMULAE

$$\sum P = 1 \quad E(X) = \sum xP \quad E(X^2) = \sum x^2P$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$E(aX + b) = aE(X) + b \quad \text{Var}(aX + b) = a^2 \text{Var}(X)$$

$$\text{Uniform on } 1..n: E = \frac{n+1}{2}, \text{Var} = \frac{n^2-1}{12}$$

- c.d.f. $F(x) = P(X \leq x)$; $P(X = x) = F(x) - F(x-1)$; last $F = 1$.
- Solve inequalities first: $3X - 3 < X + 2 \Rightarrow X < 2.5$.
- Two variables: list the pairs, multiply, add.
- $\text{Var}(5 - 2X) = 4 \text{Var}(X)$ (constant vanishes, a squared).
- Symmetric distribution: $E(X)$ is the centre ("write down").

7 NORMAL DISTRIBUTION

- $N(\mu, \sigma^2)$: 2nd parameter is **variance** ($N(443, 9) \Rightarrow \sigma = 3$).
- $Z = \frac{X-\mu}{\sigma}$; sketch and shade first.
- $P(Z < -a) = 1 - \Phi(a)$.
- $P(\mu - k < X < \mu + k) = p \Rightarrow \Phi\left(\frac{k}{\sigma}\right) = \frac{1+p}{2}$.

- Below the mean $\Rightarrow$ negative z .
- Two unknowns: two equations $\frac{x-\mu}{\sigma} = z$, subtract.
- Conditional: $\frac{P(A \cap B)}{P(B)}$ from the sketch.
- "Exactly 2 of 4": $\binom{4}{2}p^2(1-p)^2$; "at least one": $1 - (1-p)^n$.
- "700 to nearest 50" $\Rightarrow 675 \leq S < 725$.

PERCENTAGE POINTS ($P(Z > z) = p$)

p	0.40	0.30	0.25	0.20
z	0.2533	0.5244	0.6745	0.8416
p	0.15	0.10	0.05	0.025
z	1.0364	1.2816	1.6449	1.9600

8 TOP TRAPS

1. z to 2 d.p. from the percentage-points table.
2. Variance used as σ .
3. Context-free interpretation of r or gradient.
4. "Extrapolation" without quoting the x -range.
5. Independence: no conclusion stated.
6. Venn region outside the circles left blank.
7. $\text{Var}(aX + b)$ with b kept or a not squared.
8. $\sum y^2$ used as S_{yy} .
9. Wrong quartile position rule for raw vs grouped data.
10. Rounding b before finding a .

9 IN THE EXAM ROOM

EVERY QUESTION

Formula $\rightarrow$ substitution $\rightarrow$ answer (method marks sit on the substitution). Sketch every normal curve. Comments in context. About 1.2 min per mark.